



07: , , , ,

Phase 2 — | 60

Part A:

1:

$$: x^2 + 7x + 12.$$

$$\textcircled{1} 7, 12 \rightarrow 3 \cdot 4.$$

$$\textcircled{2} \rightarrow (x + 3)(x + 4).$$

$$: (x + 3)(x + 4).$$

$$x \cdot x = x^2. x \cdot 4 = 4x. 3 \cdot x = 3x. 3 \cdot 4 = 12.$$

$$: x^2 + 7x + 12.$$

$$7, -18 \rightarrow -2, 9. (x - 2)(x + 9).$$

$$-5, 6 \rightarrow -2, -3. (x - 2)(x - 3).$$

2: 1 — ac

$$2x^2 + 7x + 3.$$

$$\textcircled{1} 2 \cdot 3 \rightarrow 2 \times 3 = 6.$$

$$\textcircled{2} 7, 6 \rightarrow 6 \cdot 1.$$

$$\textcircled{3} 7x \cdot 6x + 1x \cdot x \rightarrow 2x^2 + 6x + x + 3.$$

$$\textcircled{4} \rightarrow 2x(x + 3).$$

$$\textcircled{5} \rightarrow 1(x + 3).$$

$$\textcircled{6} (x + 3)(2x + 1)(x + 3).$$

$$\diamond \diamond \diamond \diamond: 6x^2 + 5x - 6.$$

$$\textcircled{1} 6 \times (-6) = -36.$$

$$\textcircled{2} \diamond \diamond \diamond \diamond 5, \diamond \diamond \diamond \diamond -36 \rightarrow 9, -4.$$

$$\textcircled{3} \diamond \diamond \diamond \diamond: 6x^2 + 9x - 4x - 6.$$

$$\textcircled{4} \diamond \diamond \diamond \diamond: 3x(2x + 3) - 2(2x + 3).$$

$$\textcircled{5} \diamond \diamond \diamond \diamond \diamond: (3x - 2)(2x + 3).$$

$$\diamond \diamond \textbf{3:} \diamond \diamond \diamond \text{---} \diamond \diamond \diamond \diamond \diamond \diamond$$

$$x^2 + 6x + 9.$$

$$\diamond \diamond \diamond x \diamond \diamond \diamond. \diamond \diamond \diamond 3 \diamond \diamond \diamond.$$

$$\diamond \diamond \diamond 6x \diamond 2 \times x \times 3. \rightarrow (x + 3)^2 \diamond \diamond \diamond \diamond.$$

$$x^2 - 10x + 25. x \diamond \diamond \diamond, 5 \diamond \diamond \diamond, \diamond \diamond \diamond 2 \cdot x \cdot (-5) = -10x.$$

$$\rightarrow (x - 5)^2 \diamond \diamond \diamond \diamond.$$

$$4x^2 + 12x + 9. (2x)^2, 3^2, \diamond \diamond \diamond 2 \cdot 2x \cdot 3 = 12x.$$

$$\rightarrow (2x + 3)^2.$$

$$x^2 + 6x + 10. x^2, (\sqrt{10})^2, \diamond \diamond \diamond 2 \cdot x \cdot \sqrt{10} \neq 6x.$$

$$\rightarrow \diamond \diamond \diamond \diamond \diamond \diamond. \diamond \diamond \diamond \diamond \diamond.$$

$$\diamond \diamond \textbf{4:} \diamond \diamond \diamond \diamond \diamond \diamond \text{---} \diamond \diamond \diamond \diamond \diamond \diamond \diamond$$

$$x^2 - 9. x \diamond \diamond \diamond \diamond \diamond 3 \diamond \diamond \diamond.$$

$$\rightarrow (x - 3)(x + 3).$$

$$4x^2 - 25. (2x)^2 - 5^2.$$

$$\rightarrow (2x - 5)(2x + 5).$$

$$x^4 - 16. (x^2)^2 - 4^2 \rightarrow (x^2 - 4)(x^2 + 4).$$

$$x^2 - 4 \diamond \diamond \diamond \diamond \diamond: (x - 2)(x + 2).$$

$$\rightarrow \diamond \diamond: (x - 2)(x + 2)(x^2 + 4). x^2 + 4 \diamond \diamond \diamond \diamond \diamond \diamond.$$

5: —

$$x^3 - 8. x \quad 2$$

$$\rightarrow (x - 2)(x^2 + 2x + 4).$$

$$x^3 + 27. x^3 + 3^3.$$

$$\rightarrow (x + 3)(x^2 - 3x + 9).$$

$$8x^3 - 1. (2x)^3 - 1^3.$$

$$\rightarrow (2x - 1)(4x^2 + 2x + 1).$$

6: —

$$3x^3 - 12x.$$

$$\textcircled{1} 3x \rightarrow 3x(x^2 - 4).$$

$$\textcircled{2} x^2 - 4 = (x - 2)(x + 2).$$

$$\rightarrow 3x(x - 2)(x + 2).$$

$$2x^4 - 32.$$

$$\textcircled{1} 2x^4 \rightarrow 2(x^4 - 16).$$

$$\textcircled{2} (x^2 - 4)(x^2 + 4).$$

$$\textcircled{3} x^2 - 4: (x - 2)(x + 2).$$

$$\rightarrow 2(x - 2)(x + 2)(x^2 + 4).$$

Part B: 3 —

7: —

$$x^3 - 6x^2 + 11x - 6.$$

① $\pm 1, \pm 2, \pm 3, \pm 6$.

② $\pm 1, \pm 2, \pm 3, \pm 6$.

$x = 1: 1 - 6 + 11 - 6 = 0. \rightarrow (x - 1)$

③ $[1, -6, 11, -6]$

	1	-6	11	-6	
1		1	-5	6	
	1	-5	6	0	$\leftarrow 0!$

$\pm 1, \pm 2, \pm 3, \pm 6$:

- $\pm 1: 1 - 6 + 11 - 6 = 0.$
- $\pm 2: 8 - 24 + 22 - 6 = -2.$
- $\pm 3: 27 - 54 + 33 - 6 = 0.$
- $\pm 6: 216 - 216 + 66 - 6 = 60.$

④ $[1, -5, 6]$

⑤ $(x - 2)(x - 3).$

$\rightarrow (x - 1)(x - 2)(x - 3).$

8: —

$$2x^3 - 3x^2 - 3x + 2.$$

① $\pm 1, \pm 2, \pm \frac{1}{2}$.

② $x = 1: 2 - 3 - 3 + 2 = -2.$

③ $x = -1: -2 - 3 + 3 + 2 = 0. \rightarrow (x + 1)$

④ $2x^2 - 5x + 2.$

⑤ $\diamond\diamond\diamond\diamond\diamond: (2x-1)(x-2).$

$\rightarrow \diamond: x = -1, x = \frac{1}{2}, x = 2.$

$\diamond\diamond 9: t\diamond\diamond\diamond\diamond\diamond\diamond\diamond$

$x^4 - 5x^2 + 4.$

① $x^2\diamond t\diamond\diamond\diamond. \rightarrow t^2 - 5t + 4.$

② $\diamond\diamond\diamond: (t-1)(t-4).$

③ $t\diamond x^2\diamond\diamond\diamond\diamond\diamond: (x^2-1)(x^2-4).$

④ $\diamond\diamond\diamond\diamond\diamond\diamond\diamond: (x-1)(x+1)(x-2)(x+2).$

$x^6 - 9x^3 + 8.$

① $x^3\diamond t\diamond. \rightarrow t^2 - 9t + 8.$

② $(t-1)(t-8).$

③ $\diamond\diamond\diamond\diamond: (x^3-1)(x^3-8).$

④ $\diamond\diamond\diamond\diamond\diamond: (x-1)(x^2+x+1)(x-2)(x^2+2x+4).$

$\diamond\diamond 10: \diamond\diamond\diamond\diamond\diamond\diamond\diamond - x^2\diamond\diamond\diamond\diamond$

$x^4 + x^3 - 4x^2 + x + 1 = 0.$

① $\diamond\diamond\diamond\diamond: 1, 1, -4, 1, 1. \diamond\diamond\diamond\diamond\diamond\diamond\diamond.$

② $x = 0\diamond\diamond\diamond\diamond\diamond. \diamond\diamond\diamond x^2\diamond\diamond\diamond\diamond.$

$x^2 + x - 4 + \frac{1}{x} + \frac{1}{x^2} = 0.$

③ $x + \frac{1}{x}\diamond t\diamond\diamond\diamond.$

$x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2 = t^2 - 2.$

④ $\diamond\diamond: (t^2-2) + t - 4 = 0 \rightarrow t^2 + t - 6 = 0.$

⑤ $\diamond\diamond\diamond: (t+3)(t-2) = 0. \rightarrow t = -3 \diamond\diamond t = 2.$

$$\textcircled{6} t = -3: x + \frac{1}{x} = -3 \rightarrow x^2 + 3x + 1 = 0 \rightarrow x = \frac{-3 \pm \sqrt{5}}{2}.$$

$$\textcircled{7} t = 2: x + \frac{1}{x} = 2 \rightarrow (x-1)^2 = 0 \rightarrow x = 1 (\diamond\diamond, \diamond\diamond).$$

$\diamond\diamond$ 11: $\diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond \text{---} \diamond\diamond \diamond$

$$\diamond\diamond\diamond\diamond x^2 + px + q = 0 \diamond\diamond\diamond r_1, r_2 \diamond\diamond\diamond.$$

$$\diamond: r_1 + r_2 = -p. \diamond: r_1 r_2 = q.$$

$$\diamond\diamond\diamond\diamond: x^2 - 5x + 6 = 0 \rightarrow \diamond 5, \diamond 6 \rightarrow \diamond\diamond 2, 3.$$

$$\diamond\diamond\diamond\diamond 2 + \sqrt{3} \diamond\diamond\diamond\diamond 4 \diamond\diamond, \diamond\diamond\diamond\diamond 2 - \sqrt{3}. \diamond\diamond 4 \diamond\diamond.$$

$$\diamond\diamond\diamond\diamond x^3 + ax^2 + bx + c = 0 \diamond\diamond\diamond r_1, r_2, r_3:$$

- $r_1 + r_2 + r_3 = -a.$
- $r_1 r_2 + r_2 r_3 + r_3 r_1 = b.$
- $r_1 r_2 r_3 = -c.$

Part C: $\diamond\diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond$

$\diamond\diamond$ 12: $\diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond \diamond\diamond$

$$\frac{5x-1}{x^2-x-2}.$$

$$\textcircled{1} \diamond\diamond\diamond \diamond\diamond \diamond\diamond\diamond: (x-2)(x+1).$$

$$\textcircled{2} \frac{5x-1}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1} \diamond\diamond\diamond \diamond\diamond\diamond\diamond.$$

$$\textcircled{3} \diamond\diamond\diamond (x-2)(x+1) \diamond\diamond\diamond: 5x-1 = A(x+1) + B(x-2).$$

$$\textcircled{4} x = 2 \diamond\diamond\diamond\diamond. B \diamond\diamond\diamond\diamond \rightarrow 9 = 3A \rightarrow A = 3.$$

$$\textcircled{5} x = -1 \diamond\diamond\diamond\diamond. A \diamond\diamond\diamond\diamond \rightarrow -6 = -3B \rightarrow B = 2.$$

$$\rightarrow \frac{3}{x-2} + \frac{2}{x+1}.$$

$$\text{13:}$$

$$\frac{2x+1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}.$$

$$\textcircled{1} \hspace{0.2em} (x-1)^2: 2x+1 = A(x-1) + B.$$

$$\textcircled{2} \hspace{0.2em} x = 1: 3 = B.$$

$$\textcircled{3} \hspace{0.2em} x = 0: 1 = -A + 3 \rightarrow A = 2.$$

$$\rightarrow \frac{2}{x-1} + \frac{3}{(x-1)^2}.$$

$$\text{14:}$$

$$\frac{x^2}{x-1}.$$

$$\textcircled{1} \hspace{0.2em} (2)(1).$$

$$\textcircled{2} \hspace{0.2em} x^2 \div (x-1) = x + 1 + \frac{1}{x-1}.$$

$$\textcircled{3} \hspace{0.2em} \frac{1}{x-1}.$$

$$\rightarrow x + 1 + \frac{1}{x-1}.$$

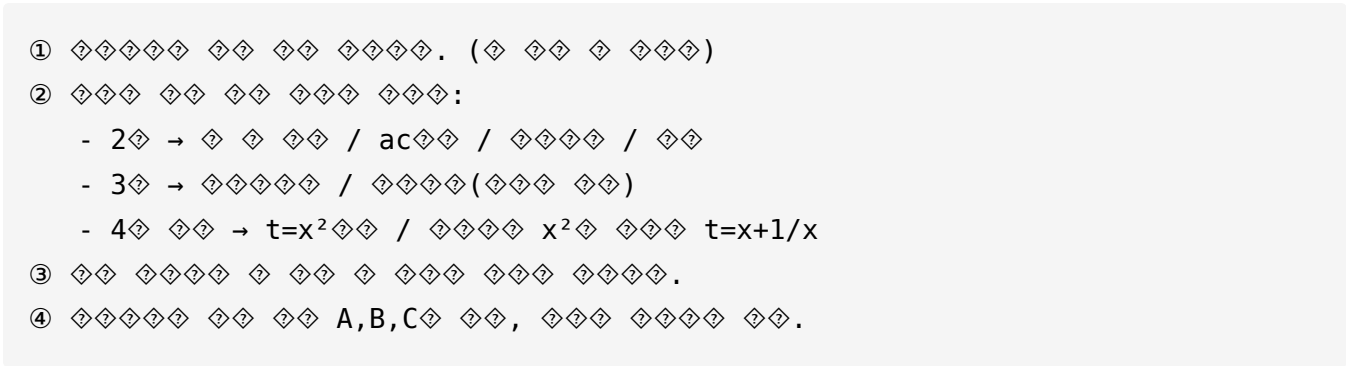
$$\text{1:}$$

$$\text{1:}$$

$$: "x = -1 \hspace{0.2em} 0. \hspace{0.2em} (x-1)."$$

$$: x = -1 \hspace{0.2em} 0 \hspace{0.2em} x - (-1) = x + 1.$$

$$.$$



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→ $\diamondsuit ? \diamondsuit ? \diamondsuit ? \diamondsuit ? : \diamondsuit ? \diamondsuit ?$ 12, 13

???: ???

?? 4: ???

??? 7, ??? 12 ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? 3 ? ? ? ? .
? ? ? 1 ? ? 1 ? , ? ? ? 1 ? ? ? 2 ? .

→ ? ? ? ? : ? ? 1, 2

???: ???

?? 5

$x^4 - 2x^3 - 13x^2 - 2x + 1 = 0$? ? ? ? ? ? ? ? . (? ? ? ? → x^2 ? ? ? ? ?)

→ ? ? ? ? : ? ? 10

???: ???

?? 6: ??

$x^3 - 3x^2 + ax + b = 0$? ? ? ? 1, r , r^2 ? ? ? . a, b, r ? ? ? ? ? ? .
? ? ? : ? = 3, ? ? ? ? ? = a, ? ? = -b.

→ ? ? ? ? : ? ? 11

???: ???

?? ?? ??

1. : . 2.
2. :
- 2: / ac / /
- 3: / ()
- 4: $t=x^2$ / $x^2 \rightarrow x^2$ $t=x+1/x$
- : A,B,C
3. : .

?? ??

" ", " ", " ", " ", " " .

		/
		factor
		expand
		factor out
		$(a \pm b)^2 = a^2 \pm 2ab + b^2$
		$a^2 - b^2 = (a - b)(a + b)$
		$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
	synthetic division	$(x - r)$
		Vieta's formulas
		reciprocal equation
		partial fraction
		multiple root